Supplementary Information

Investigating the Relationship Between AI-Based Structure Prediction Confidence and Experimentally Determined Protein Stability

Paper BJ-1129 · Revised 7 November 2026

Scope and data definitions

This supplement records the 20 canonical WT proteins, all 200 experimental mutation labels, additional model summaries and the analyses added during revision. The experimental endpoint is folding ΔΔG from Pak et al., Supplementary Table S1, derived from ThermoMutDB. Positive values denote destabilisation and negative values stabilisation. Exact-zero values are retained. Historical experimental constructs were not independently reconstructed.

Each protein contributes ten substitutions sharing a matched WT baseline. Local ΔpLDDT is MT minus WT at the mutated residue; global ΔpLDDT is the difference between equal-residue sequence means. Explicit native confidence arrays on a 0–1 scale were multiplied by 100. Three signed seed-label differences are averaged equally for each mutation and model. Seed labels 42, 43 and 44 are computational repeats, not extra experimental observations.

This self-contained appendix presents the supplementary tables and figures cited in the main manuscript. It includes the complete experimental-label inventory and additional reported results. Raw prediction outputs, intermediate processing files, executable analysis scripts and separate machine-readable exports are not part of this appendix release.

Analysis added during revision

The signed Kyte–Doolittle hydrophobicity change is H(mutant residue) − H(WT residue). The BLOSUM62 baseline is the untransformed matrix entry for that substitution, in conventional half-bit units. Neither baseline is trained, sign-selected to maximise agreement, or calibrated as a folding energy. Both use only mutation identities. Kyte–Doolittle values are represented as integer tenths before subtraction; exact-tenth differences are used before ranking so binary floating-point noise cannot split genuine ties.

Pooled 95% intervals use 10,000 protein-cluster percentile bootstrap draws. Twenty accessions are drawn with replacement in each replicate, preserving all ten mutations for every selected cluster. Baseline calculations use NumPy default_rng(20260916), lexicographically sorted accessions and saved shared draws. Pairwise comparisons require both features on those same rows and draws. Baseline calculations independently recovered every experimental label from the submitted Appendix A2 and reproduced the reported count, range, moments and quartiles.

Per-protein Pearson Fisher-z intervals, where reported, are approximate pointwise intervals with n = 10. They assume independent bivariate-normal mutation pairs conditional on the shared predicted WT. They omit uncertainty in that shared WT baseline and experimental measurement error, and are not multiplicity adjusted. Their purpose is to display heterogeneity, not to establish model superiority for an individual protein.

Figure S1 Workflow

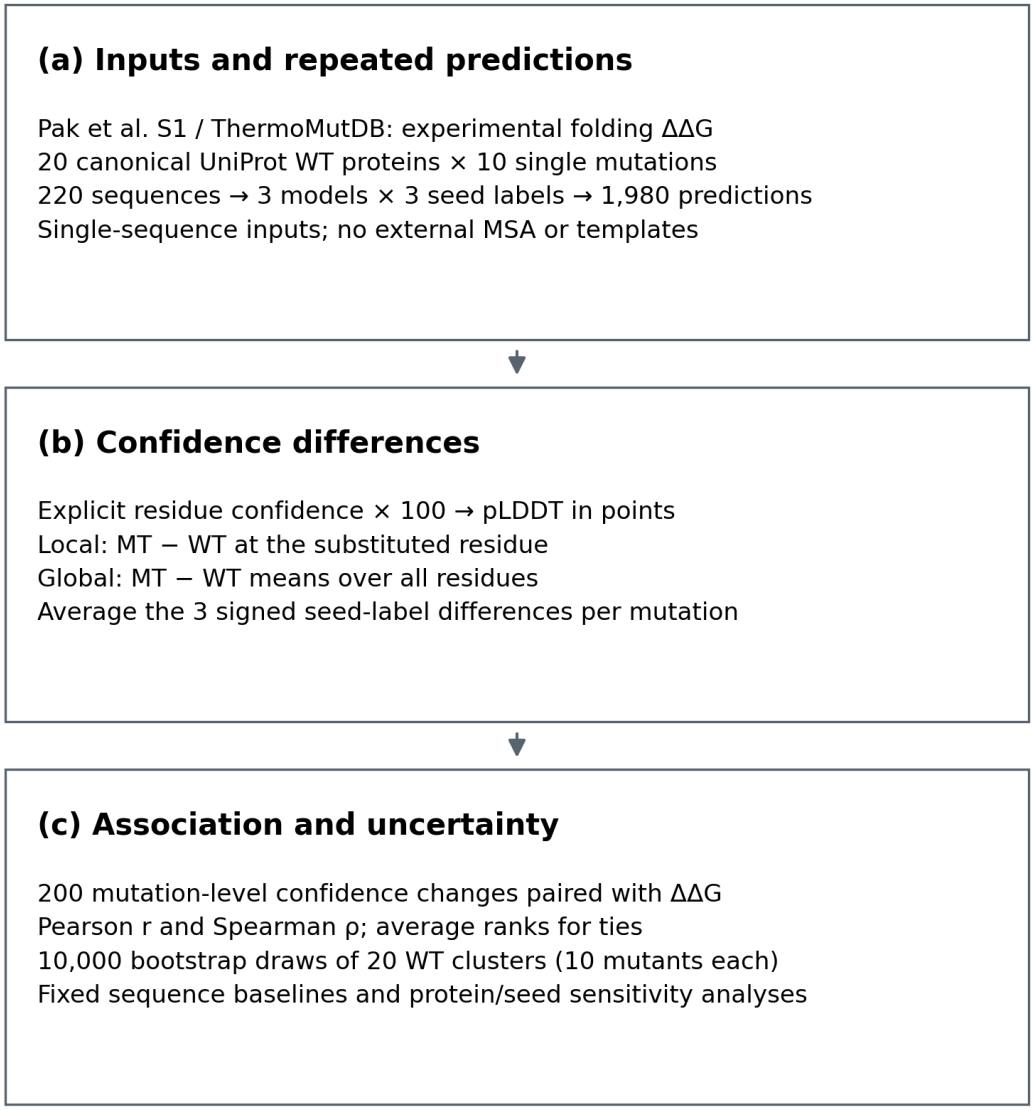


Figure S1. (a) Experimental labels and canonical sequences define the shared prediction panel. (b) Signed local and global confidence differences are averaged across three seed labels. (c) Association, cluster uncertainty and sensitivity analyses are evaluated at the mutation level. Shared seed labels do not guarantee identical WT–MT random-number trajectories.

Table S1 Canonical WT protein inventory

Sequences were retrieved from reviewed UniProtKB/Swiss-Prot entries on 15 September 2026. Every accession contributed ten single substitutions. Length refers to the complete canonical input sequence, not necessarily the construct used in a historical assay.

UniProt accession	Protein	Length (residues)
P00044	Cytochrome c isoform 1	109
P00149	Cytochrome c′	146
P00651	Guanyl-specific ribonuclease T1	130
P00711	Alpha-lactalbumin	142
P00720	Endolysin	164
P00974	Pancreatic trypsin inhibitor	100
P01053	Subtilisin-chymotrypsin inhibitor-2A	84
P02185	Myoglobin	154
P02417	50S ribosomal protein L9	149
P02633	Protein S100-G	79
P03050	Transcriptional repressor arc	53
P05230	Fibroblast growth factor 1	155
P05798	Guanyl-specific ribonuclease Sa	96
P0A3E0	Flavodoxin	170
P0A7Y4	Ribonuclease HI	155
P23370	30S ribosomal protein S6	101
P29160	50S ribosomal protein L30e	101
P41016	Cold shock protein CspB	66
P62942	Peptidyl-prolyl cis-trans isomerase FKBP1A	108
P69543	DNA-binding protein G5P	87

Table S2 Local and global correlation summaries

Pointwise 95% protein-cluster intervals below preserve the manuscript results. All six rows use the same 200 observations; intervals for different features are not themselves tests of differences.

Model and endpoint	Spearman ρ [95% CI]	Pearson r [95% CI]
Boltz-2 local	−0.106 [−0.278, 0.076]	0.038 [−0.168, 0.198]
Boltz-2 global	−0.078 [−0.245, 0.117]	0.063 [−0.144, 0.193]
Chai-1 local	−0.271 [−0.408, −0.122]	−0.235 [−0.394, −0.095]
Chai-1 global	−0.287 [−0.416, −0.132]	−0.227 [−0.372, −0.111]
ESMFold2 local	−0.255 [−0.414, −0.109]	−0.218 [−0.388, −0.060]
ESMFold2 global	−0.281 [−0.422, −0.139]	−0.134 [−0.413, 0.077]

Table S3 Local confidence changes by stability direction

Values are median [first quartile, third quartile] in pLDDT points. The groups contain 43 stabilising and 153 destabilising substitutions. The four zero-valued observations remain in every pooled correlation and are excluded only from this two-group display.

Model	Stabilizing ΔΔG < 0 n = 43	Destabilizing ΔΔG > 0 n = 153
Boltz-2	−1.215 [−5.827, 1.943]	−2.321 [−8.734, 1.307]
Chai-1	0.384 [−1.412, 3.819]	−0.364 [−3.459, 0.939]
ESMFold2	0.362 [−1.443, 4.013]	−0.144 [−1.840, 0.519]

Table S4 Fixed baseline correlations

The baselines were evaluated on exactly the same experimental labels and do not use model outputs. Point estimates and intervals are shown to three decimal places.

Feature	Spearman ρ [95% CI]	Pearson r [95% CI]
Signed hydrophobicity change	−0.230 [−0.387, −0.078]	−0.193 [−0.375, −0.022]
BLOSUM62 substitution score	0.006 [−0.190, 0.178]	−0.061 [−0.224, 0.086]

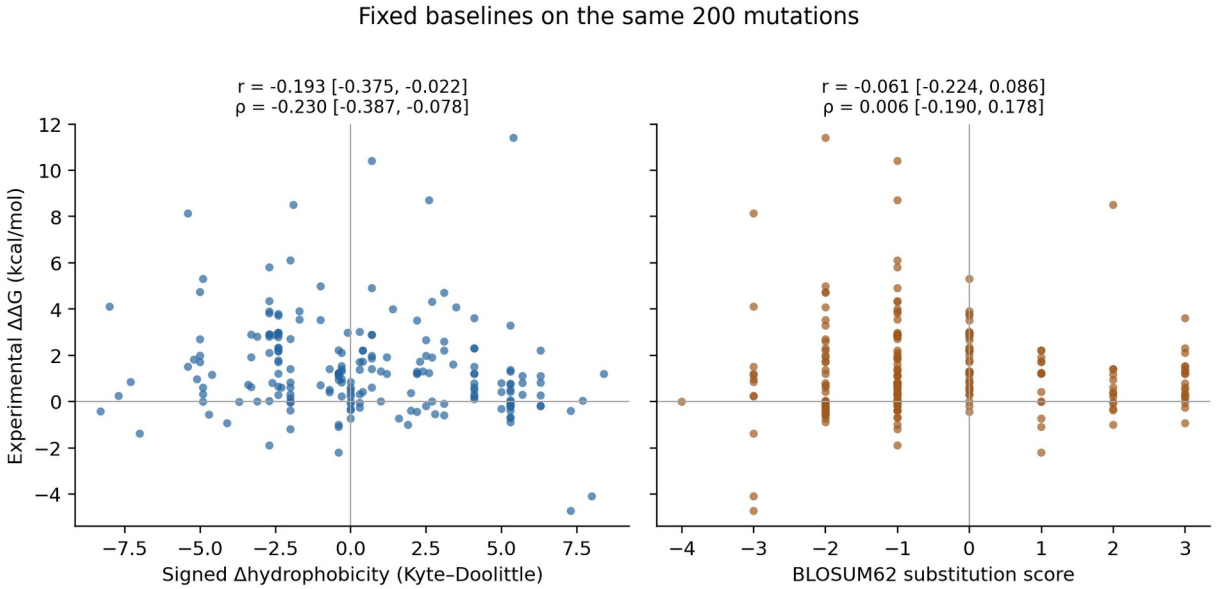


Figure S2. Fixed sequence baselines versus experimental folding ΔΔG. Hydrophobicity is mutant minus WT; BLOSUM62 is the raw substitution score. Each panel retains all 200 observations. The displayed features are not fitted energy predictions.

Table S5 Complete experimental mutation records

Mutation IDs encode accession, WT residue, one-based canonical position and mutant residue. ThermoMutDB and S1 row identifiers retain the original source links. Values retain Pak et al.'s sign convention, including exact zeros; no sign reversal was applied.

Mutation ID	ΔΔG (kcal/mol)	ThermoMutDB ID	Pak S1 row	WT accession
P00044_G12A	3.5	2016	432	P00044
P00044_K28H	10.4	6640	914	P00044
P00044_K33H	4.9	6639	913	P00044
P00044_K33Q	1.72	6634	912	P00044
P00044_N58H	1.7	2009	431	P00044
P00044_N58I	-4.1	1982	427	P00044
P00044_N58L	-4.73	1981	426	P00044
P00044_N58Q	0.84	2006	430	P00044
P00044_K78I	1.19	2248	448	P00044
P00044_K78L	0.03	2244	447	P00044
P00149_D24H	1.37	9511	1103	P00149
P00149_K34H	0.6	9510	1102	P00149
P00149_K41H	2.88	9509	1101	P00149
P00149_K60H	1.86	9507	1100	P00149
P00149_A87H	1.98	9504	1099	P00149
P00149_E94H	3.01	9503	1098	P00149
P00149_K105H	2.88	9502	1097	P00149
P00149_A112H	1.7	9501	1096	P00149
P00149_K118H	1.97	9500	1095	P00149
P00149_A125H	2.69	9499	1094	P00149
P00651_N35A	-0.9	665	162	P00651
P00651_S38A	1.23	662	159	P00651
P00651_V42A	2.19	5741	778	P00651
P00651_G49A	1.2	4899	720	P00651
P00651_N70S	1.9	4157	661	P00651
P00651_Y82F	0.78	659	157	P00651
P00651_Y83F	0.5	660	158	P00651
P00651_W85Y	0.93	2414	457	P00651
P00651_V104S	4.73	5750	782	P00651
P00651_V115C	3.54	5744	779	P00651
P00711_V61G	1.15	4542	698	P00711
P00711_V61N	0.24	4540	697	P00711
P00711_Q73A	0.41	4543	699	P00711
P00711_Y122P	0.22	4545	701	P00711
P00711_H126A	0.79	4535	692	P00711
P00711_H126W	1.72	4536	693	P00711
P00711_L129H	-1.39	4527	685	P00711
P00711_L129R	-0.43	4528	686	P00711
P00711_W137H	0.6	4532	690	P00711
P00711_W137Y	1.17	4531	689	P00711
P00720_I50A	2.9	8691	1078	P00720
P00720_G77A	1.2	874	179	P00720
P00720_A82P	0.72	875	180	P00720

Mutation ID	ΔΔG (kcal/mol)	ThermoMutDB ID	Pak S1 row	WT accession
P00720_P86G	1.91	876	181	P00720
P00720_L91P	8.13	877	182	P00720
P00720_R119E	0.0	5355	749	P00720
P00720_A146I	4.31	884	185	P00720
P00720_A146T	2.87	885	186	P00720
P00720_I150V	2.1	8696	1082	P00720
P00720_T157F	4.07	891	191	P00720
P00974_T46A	-0.2	3317	585	P00974
P00974_T46M	8.7	6191	859	P00974
P00974_P48A	1.6	1881	408	P00974
P00974_A51V	1.3	1884	410	P00974
P00974_L64M	8.5	6190	858	P00974
P00974_F68L	1.3	1886	411	P00974
P00974_Y70A	4.7	1887	412	P00974
P00974_G71D	2.8	1893	416	P00974
P00974_R74A	-0.2	3318	586	P00974
P00974_E84M	11.4	6193	861	P00974
P01053_K22A	0.8	1528	292	P01053
P01053_E27Q	0.6	1533	295	P01053
P01053_E34N	0.49	1539	300	P01053
P01053_E34Q	0.34	1537	298	P01053
P01053_V39A	0.28	1548	308	P01053
P01053_I49A	3.89	2508	464	P01053
P01053_V54A	0.64	1937	421	P01053
P01053_F70A	3.52	1573	326	P01053
P01053_D72A	3.28	1574	327	P01053
P01053_I77A	4.34	2514	465	P01053
P02185_E5A	-0.7	4519	682	P02185
P02185_L12A	0.6	4338	667	P02185
P02185_E19A	-0.7	4518	681	P02185
P02185_L30A	-0.39	6867	925	P02185
P02185_I31A	-1.9	6866	924	P02185
P02185_D45A	-0.2	4521	683	P02185
P02185_D61A	-0.15	4522	684	P02185
P02185_F107A	0.7	4339	668	P02185
P02185_L116A	1.4	4343	669	P02185
P02185_A145L	-0.4	4327	666	P02185
P02417_V3I	0.31	7840	1024	P02417
P02417_V3L	2.21	7846	1028	P02417
P02417_F5V	3.99	8248	1072	P02417
P02417_L6V	0.41	7842	1026	P02417
P02417_E17Q	0.21	7833	1021	P02417
P02417_V21A	2.33	7848	1030	P02417
P02417_V21L	1.23	7844	1027	P02417
P02417_E54Q	-0.33	7829	1018	P02417
P02417_H134Q	0.97	7453	990	P02417
P02417_H144Q	0.82	7452	989	P02417

Mutation ID	ΔΔG (kcal/mol)	ThermoMutDB ID	Pak S1 row	WT accession
P02633_L10V	2.2	3974	656	P02633
P02633_Y17F	1.19	1416	282	P02633
P02633_E21Q	-0.38	1007	206	P02633
P02633_D23N	-0.74	1008	207	P02633
P02633_P24G	1.19	1415	281	P02633
P02633_E30Q	-0.09	1009	208	P02633
P02633_V65A	3.78	3979	657	P02633
P02633_F70A	4.98	3982	658	P02633
P02633_V74L	1.2	3983	659	P02633
P02633_I77V	1.53	3984	660	P02633
P03050_E17A	0.5	13289	1140	P03050
P03050_L19A	0.82	6538	875	P03050
P03050_R23A	1.1	13299	1147	P03050
P03050_E27A	-0.4	13282	1137	P03050
P03050_G30A	1.29	6539	876	P03050
P03050_V33A	1.77	6542	879	P03050
P03050_N34A	0.0	13295	1144	P03050
P03050_V41A	3.71	6547	884	P03050
P03050_M42A	2.97	6544	881	P03050
P03050_E48G	2.59	5277	739	P03050
P05230_C31S	2.89	5239	730	P05230
P05230_H36Y	-1.01	7225	972	P05230
P05230_V46W	0.96	9580	1106	P05230
P05230_L59W	-0.57	9573	1104	P05230
P05230_C98S	1.91	5233	727	P05230
P05230_H108G	-0.55	5236	729	P05230
P05230_F123Y	-0.94	7228	973	P05230
P05230_V124I	-0.06	7229	974	P05230
P05230_C132I	0.36	9578	1105	P05230
P05230_C132S	0.62	5234	728	P05230
P05798_D17K	1.1	5359	753	P05798
P05798_E41K	1.2	5357	751	P05798
P05798_Y49F	0.2	5765	788	P05798
P05798_Y51F	2.3	5766	789	P05798
P05798_Y52F	3.6	5767	790	P05798
P05798_Y55F	0.6	5768	791	P05798
P05798_E74K	-1.1	5358	752	P05798
P05798_Y80F	1.5	5762	785	P05798
P05798_Y81F	1.2	5763	786	P05798
P05798_Y86F	0.3	5764	787	P05798
P0A3E0_V19I	-0.27	7949	1059	P0A3E0
P0A3E0_I22A	0.02	7932	1052	P0A3E0
P0A3E0_I22G	-0.01	7934	1053	P0A3E0
P0A3E0_E41K	-2.21	7313	975	P0A3E0
P0A3E0_L51A	-0.05	7936	1054	P0A3E0
P0A3E0_I53V	1.38	7340	982	P0A3E0
P0A3E0_G69A	-0.45	7960	1060	P0A3E0

Mutation ID	ΔΔG (kcal/mol)	ThermoMutDB ID	Pak S1 row	WT accession
P0A3E0_Q112G	-0.1	7939	1056	P0A3E0
P0A3E0_D127K	-1.01	7316	977	P0A3E0
P0A3E0_L144A	0.14	7350	987	P0A3E0
P0A7Y4_R27A	2.2	4586	708	P0A7Y4
P0A7Y4_I53A	2.1	4584	706	P0A7Y4
P0A7Y4_I53D	4.1	6994	944	P0A7Y4
P0A7Y4_I53F	3.9	6993	943	P0A7Y4
P0A7Y4_I53L	1.4	6992	942	P0A7Y4
P0A7Y4_H62P	-0.74	44	5	P0A7Y4
P0A7Y4_S68G	2.2	28	2	P0A7Y4
P0A7Y4_Q105G	2.2	4585	707	P0A7Y4
P0A7Y4_H124A	-0.32	604	150	P0A7Y4
P0A7Y4_H127A	0.41	605	151	P0A7Y4
P23370_V6A	1.7	7702	999	P23370
P23370_I8A	2.9	7703	1000	P23370
P23370_L21A	-1.2	7704	1001	P23370
P23370_I26A	2.8	7706	1002	P23370
P23370_Y33A	-0.6	7707	1003	P23370
P23370_L48A	0.3	7710	1005	P23370
P23370_V65A	2.3	7713	1008	P23370
P23370_L79A	2.7	7715	1009	P23370
P23370_V85A	2.9	7716	1010	P23370
P23370_V88A	1.2	7717	1011	P23370
P29160_E7A	0.77	7907	1036	P29160
P29160_K10A	0.29	7909	1038	P29160
P29160_D13A	-0.24	7910	1039	P29160
P29160_K47A	1.1	7919	1043	P29160
P29160_E48A	0.05	7920	1044	P29160
P29160_R55A	0.26	7923	1045	P29160
P29160_E63A	1.29	7924	1046	P29160
P29160_E70A	0.74	7926	1048	P29160
P29160_R77A	-0.19	7927	1049	P29160
P29160_D88A	1.36	7929	1050	P29160
P41016_Q2L	-0.41	6126	839	P41016
P41016_N11S	0.0	6127	840	P41016
P41016_Y15F	0.1	6115	836	P41016
P41016_G23Q	0.0	6131	843	P41016
P41016_F38W	-0.02	7009	945	P41016
P41016_E46A	-0.1	6134	845	P41016
P41016_Q53E	-0.02	6125	838	P41016
P41016_V64T	0.31	6117	837	P41016
P41016_L66A	-0.02	6128	841	P41016
P41016_L66E	0.84	6129	842	P41016
P62942_V5A	2.78	3772	608	P62942
P62942_V24A	2.97	4554	703	P62942
P62942_T28A	1.97	3802	624	P62942
P62942_I57T	1.81	3780	612	P62942

Mutation ID	ΔΔG (kcal/mol)	ThermoMutDB ID	Pak S1 row	WT accession
P62942_R58A	0.81	3790	618	P62942
P62942_R58G	2.29	3791	619	P62942
P62942_V64A	2.97	3781	613	P62942
P62942_T76A	2.65	3804	626	P62942
P62942_I77A	3.81	3783	614	P62942
P62942_V99A	2.16	3787	615	P62942
P69543_V19T	0.6	1698	352	P69543
P69543_C33A	0.5	1666	331	P69543
P69543_V35T	5.3	1671	333	P69543
P69543_L37I	1.4	1674	336	P69543
P69543_I47A	5.8	930	195	P69543
P69543_I47L	0.4	933	198	P69543
P69543_L49A	6.1	1681	340	P69543
P69543_L65P	1.5	2709	485	P69543
P69543_M77T	0.8	1726	364	P69543
P69543_I78V	1.3	1693	349	P69543

Software and inference provenance

The successful Boltz freeze verifies boltz 2.2.1, PyTorch 2.7.1, pytorch-lightning 2.5.0 and NumPy 1.26.4. Initial Chai and ESM version-capture files contain “No module named pip”, not valid package manifests. An exact Chai version and immutable ESM source revision are therefore not established by those captures. Boltz-2 used three recycles and 200 sampling steps; Chai-1 used three trunk recycles and 200 diffusion timesteps; Biohub ESMFold2 used 20 loops and 200 sampling steps. These model-specific operations are not equivalent compute budgets. Each sequence used labels 42, 43 and 44 with one sample per label. Chai-1 and ESMFold2 reset seeds per sequence, whereas Boltz-2 seeded each batch.

Boltz-2 official implementation and output documentation: https://github.com/jwohlwend/boltz and https://github.com/jwohlwend/boltz/blob/main/docs/prediction.md. Chai-1 implementation: https://github.com/chaidiscovery/chai-lab. Biohub ESMFold2 model card: https://huggingface.co/biohub/ESMFold2. Package versions do not uniquely establish checkpoint revisions. The ESMFold2 implementation is distinct from the 2023 ESMFold model.

Biohub ESMFold2 used the full local checkpoint, local ESMC-6B embeddings and recorded language-model dropout of 0.3; its successful environment used PyTorch 2.11.0+cu128. Boltz-2 had an empty MSA and optional optimised kernels disabled. Chai-1 used ESM embeddings with MSA and template services disabled. No partners, restraints, affinity outputs or computational folding-energy estimates entered the confidence analysis.

S6 Recorded execution costs and environment

Archived host wall-clock timers were recorded for all 660 Chai-1 and 660 successful ESMFold2 predictions and are summarised in Table S6. They begin before per-item input preparation and include the model call, structure/output writing, confidence conversion and sequence validation. They end before writing the completion JSON. ESM model/ESMC loading occurs before the loop and is excluded. The Chai mean retains first-call overhead; its first item is also its maximum. These are host timers, not CUDA-event kernel measurements.

Table S6. Recorded execution time by model and seed. Per-item results show mean ± sample SD and range in seconds; Boltz has no comparable per-item timer.

Model	Seed	Items	Mean ± SD (s)	Range (s)
Chai-1	all	660	32.398 ± 3.784	29.533–77.073
Chai-1	42	220	32.304 ± 4.358	29.533–77.073
Chai-1	43	220	32.632 ± 3.790	29.755–46.390
Chai-1	44	220	32.258 ± 3.106	29.670–40.674
ESMFold2	all	660	5.110 ± 1.549	3.047–8.608
ESMFold2	42	220	5.116 ± 1.552	3.056–8.220
ESMFold2	43	220	5.107 ± 1.550	3.051–8.167
ESMFold2	44	220	5.107 ± 1.552	3.047–8.608

Boltz-2 final batch prediction-progress displays were 29:08, 29:03 and 29:06 for seeds 42, 43 and 44, respectively: 1,748, 1,743 and 1,746 seconds for 220 items each. The displays have seconds-level precision and do not measure individual items or full process runtime. Each batch reports zero failures. The Chai/ESM completion JSON records match the corresponding inference-log records exactly.

Boltz and Chai were launched concurrently on the selected CUDA device. ESM was attempted after both finished, failed during model loading and subsequently completed a separate retry after its PyTorch CUDA environment was repaired. The successful retry used PyTorch 2.11.0+cu128. The observed device name is NVIDIA vGPU-32GB; the underlying physical GPU, exclusive allocation and per-model FLOPs are not established. Thus, the recorded times are descriptive protocol costs, not a fair speed ranking.

The complete recorded launcher-to-final-retry envelope was 26,502 seconds (7 h 21 min 42 s), including concurrency, failed loading and unpartitioned waiting/repair. It is not a sum of GPU-hours or billed time. Device-wide ten-second monitoring recorded sampled maxima of 10,670 MiB during overlapping Boltz/Chai and 15,141 MiB during the ESM retry. Those samples cannot attribute memory to individual concurrent models, establish isolated peaks, or exclude shorter unsampled peaks. No energy or monetary cost is inferred.

S7 Paired baseline comparisons

Differences below are signed model coefficients minus baseline coefficients, recalculated in the same 10,000 cluster draws. They are not absolute-correlation differences, predictive-error measures, or evidence of independent predictive information. All local Spearman intervals against signed hydrophobicity include zero, without proving equivalence. Only the Chai-1 versus BLOSUM62 Spearman pointwise interval excludes zero; no multiplicity correction was applied.

Table S7. Paired model-minus-baseline local coefficient differences.

Model	Baseline	Metric	Difference [95% CI]
Boltz-2	Hydrophobicity	Pearson r	0.231 [-0.026, 0.483]
Boltz-2	BLOSUM62	Pearson r	0.099 [-0.066, 0.240]
Boltz-2	Hydrophobicity	Spearman ρ	0.123 [-0.104, 0.377]
Boltz-2	BLOSUM62	Spearman ρ	-0.112 [-0.290, 0.102]
Chai-1	Hydrophobicity	Pearson r	-0.042 [-0.237, 0.166]
Chai-1	BLOSUM62	Pearson r	-0.174 [-0.333, 0.001]
Chai-1	Hydrophobicity	Spearman ρ	-0.041 [-0.204, 0.126]
Chai-1	BLOSUM62	Spearman ρ	-0.277 [-0.463, -0.061]
ESMFold2	Hydrophobicity	Pearson r	-0.025 [-0.140, 0.122]
ESMFold2	BLOSUM62	Pearson r	-0.157 [-0.389, 0.061]
ESMFold2	Hydrophobicity	Spearman ρ	-0.025 [-0.198, 0.153]
ESMFold2	BLOSUM62	Spearman ρ	-0.261 [-0.510, 0.001]

S8 Per-protein heterogeneity

Each coefficient uses all ten mutations within its protein and the mean of three signed MT–WT confidence differences. Local Pearson ranges are −0.834 to 0.641 (Boltz-2), −0.959 to 0.557 (Chai-1) and −0.891 to 0.629 (ESMFold2). Local Spearman ranges are −0.830 to 0.648, −0.891 to 0.673 and −0.924 to 0.127, respectively. Negative local Spearman coefficients occur in 14/20, 17/20 and 14/20 proteins. These signs do not constitute individual-protein significance tests.

For the descriptive Pearson intervals, $z = \operatorname{arctanh}(r)$, $SE(z) = 1/\sqrt{(10-3)}$, and bounds are $\tanh[z \pm 1.95996398454 \times SE(z)]$. The conditional independence and bivariate-normal assumptions are approximations. Shared prediction uncertainty, assay error, construct mismatch and mutation-selection uncertainty are not propagated. The intervals are pointwise, not simultaneous or multiplicity adjusted. Figure S3 displays all local Pearson estimates and intervals; Figure S4 reports local Pearson and Spearman coefficients for each protein and model.

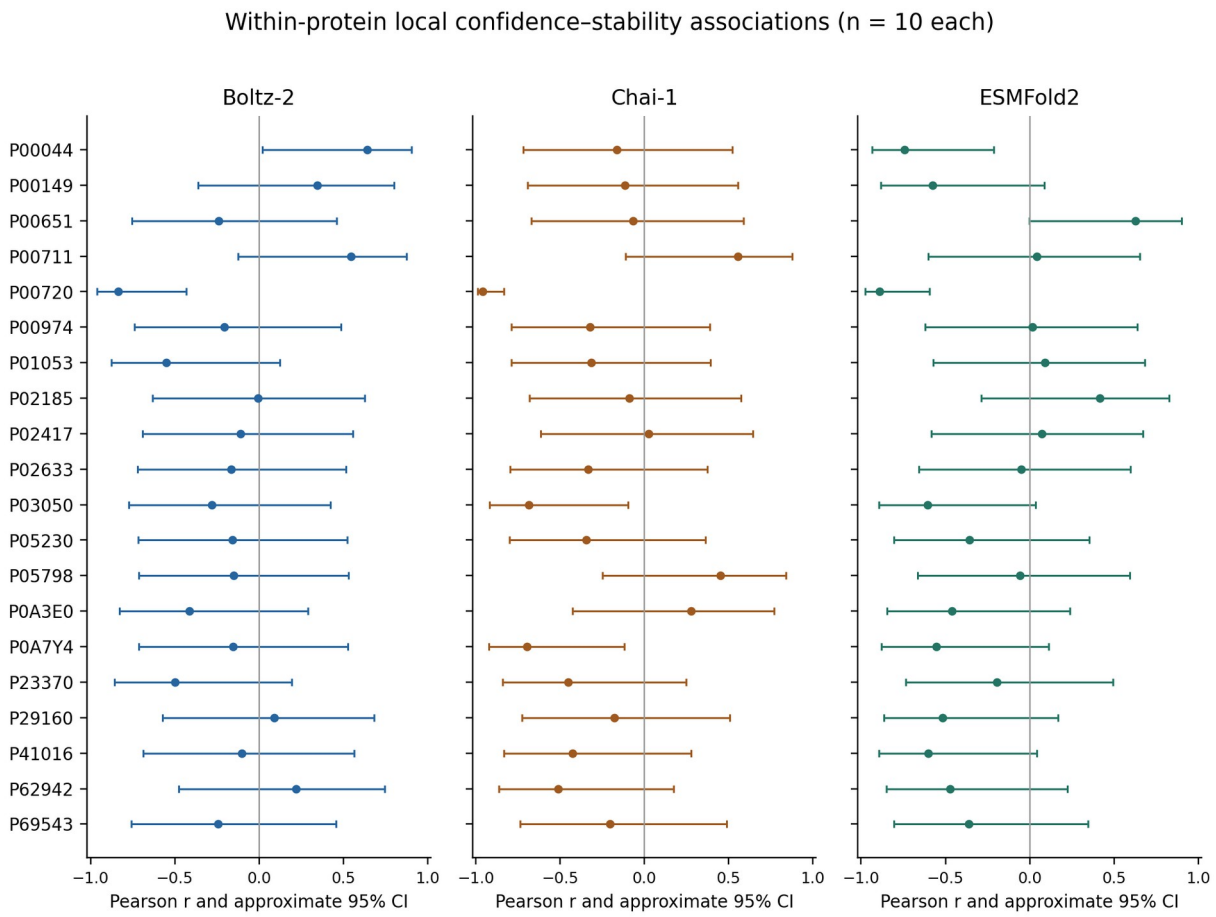


Figure S3. Local within-protein Pearson r and approximate Fisher-z 95% intervals for all 20 proteins. Every estimate uses ten mutations. All panels share the −1 to +1 correlation scale. Intervals are descriptive under the assumptions in Section S8, not model-superiority tests.

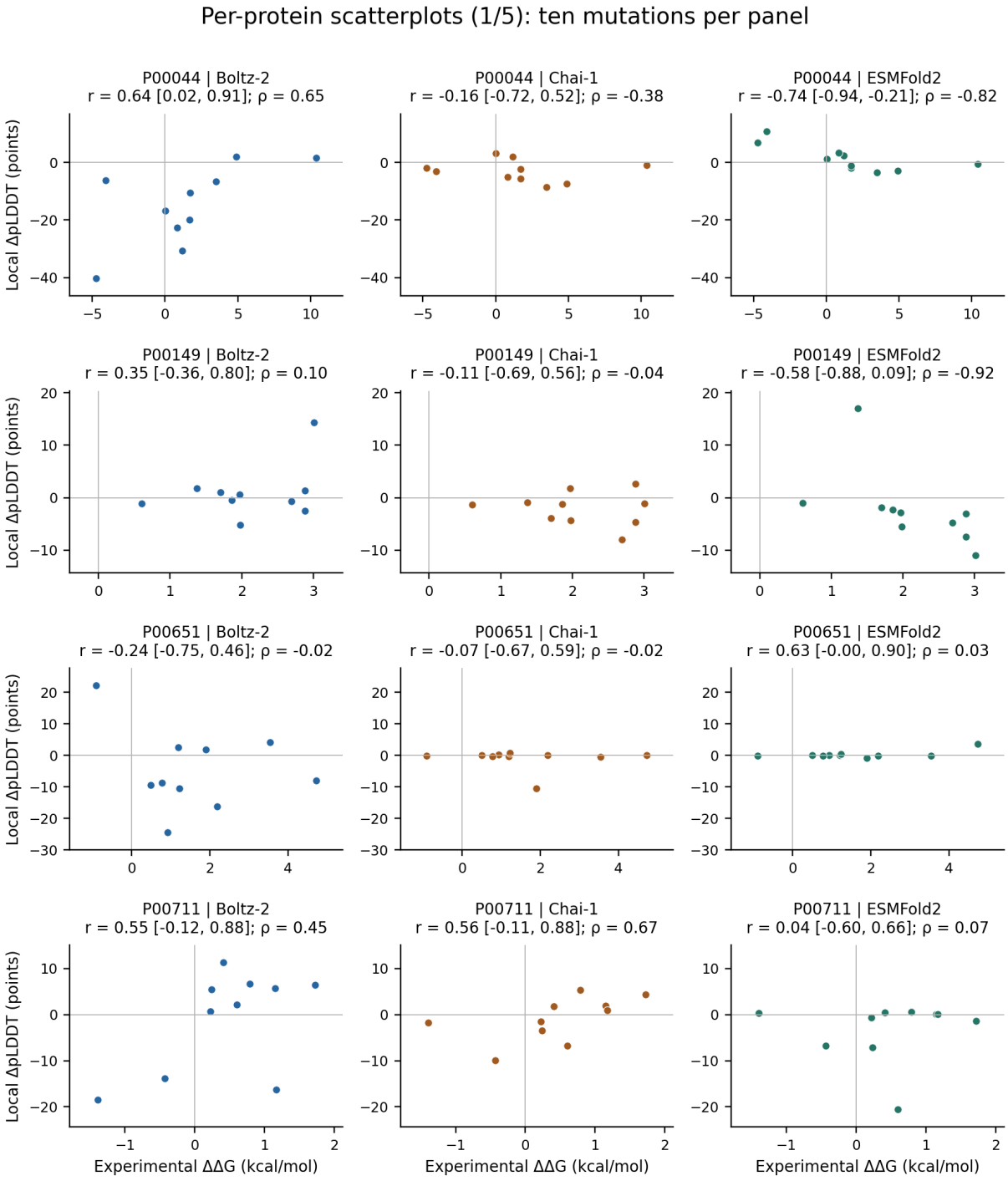


Figure S4a. Local confidence–stability scatter plots, proteins 1–4 of 20. Each panel contains all ten mutations and reports Pearson r with its approximate 95% interval and Spearman ρ. Axis ranges differ between protein rows but match across models within each row. No jitter, filtering or fitted line is applied. See Section S8 for interval assumptions.

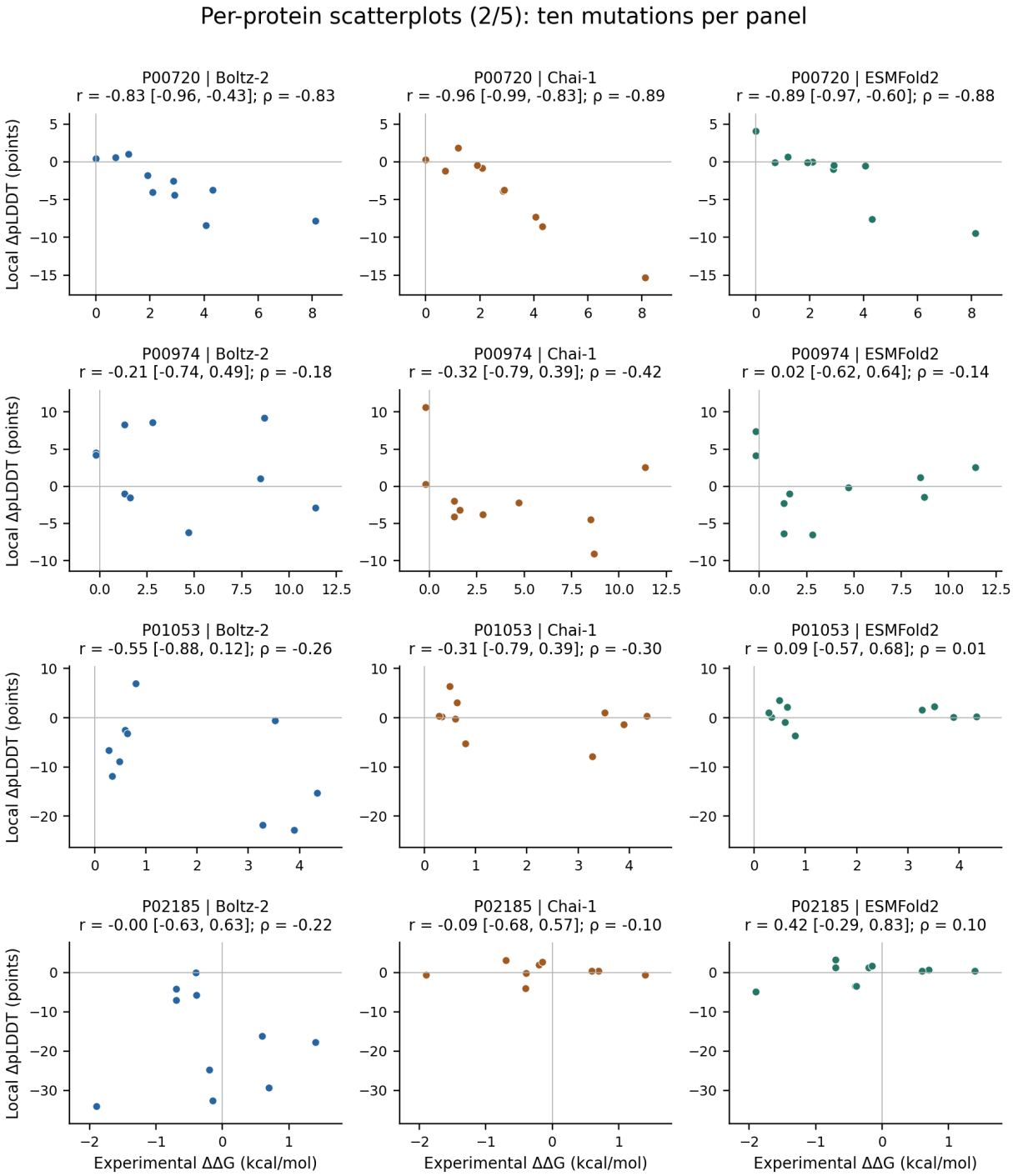
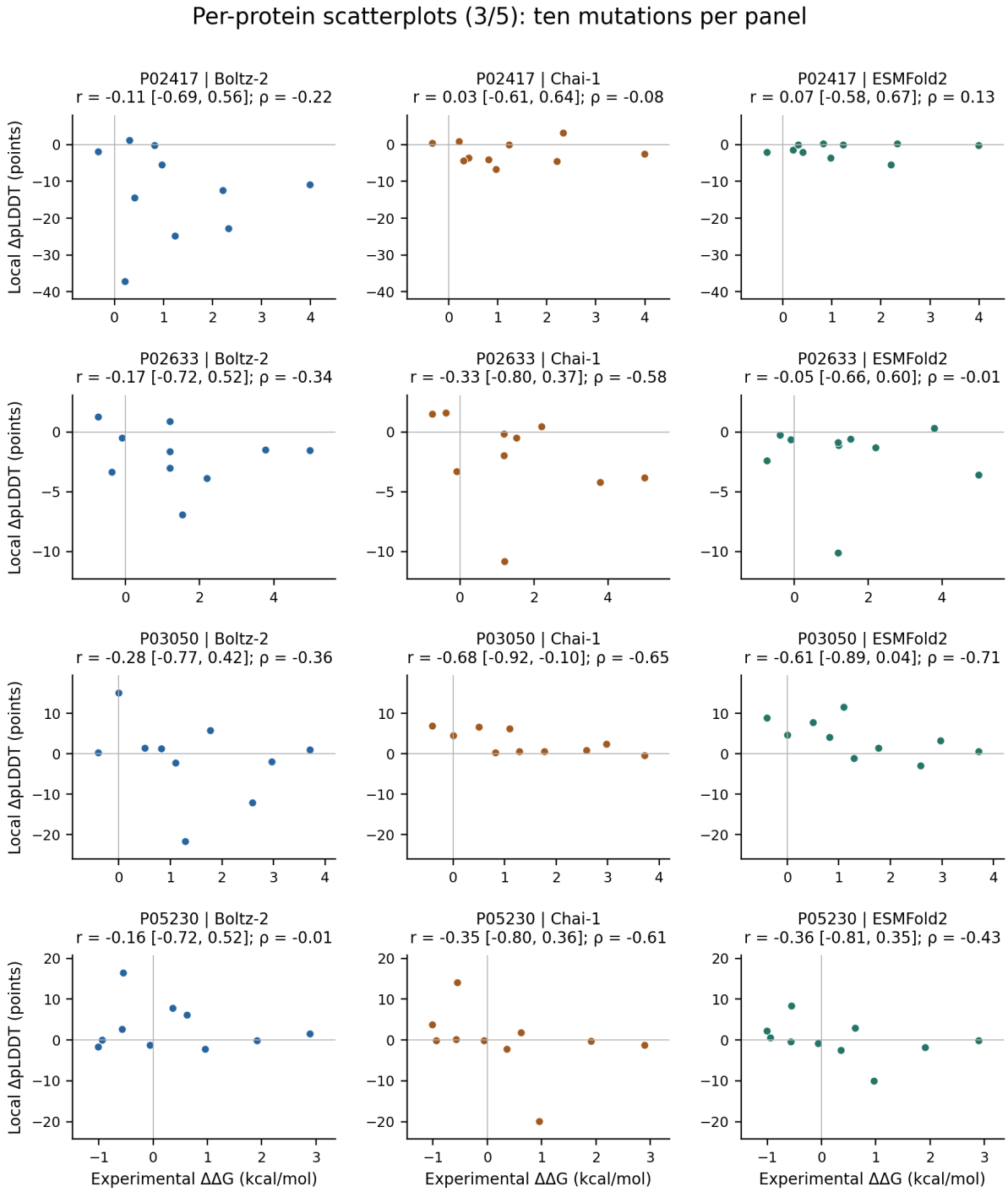


Figure S4b. Local confidence–stability scatter plots, proteins 5–8 of 20. Each panel contains all ten mutations and reports Pearson r with its approximate 95% interval and Spearman ρ. Axis ranges differ between protein rows but match across models within each row. No jitter, filtering or fitted line is applied. See Section S8 for interval assumptions.



Scales vary by protein; three model panels share scales within each row. Intervals are descriptive Fisher-z 95% CIs.

Figure S4c. Local confidence–stability scatter plots, proteins 9–12 of 20. Each panel contains all ten mutations and reports Pearson r with its approximate 95% interval and Spearman ρ. Axis ranges differ between protein rows but match across models within each row. No jitter, filtering or fitted line is applied. See Section S8 for interval assumptions.

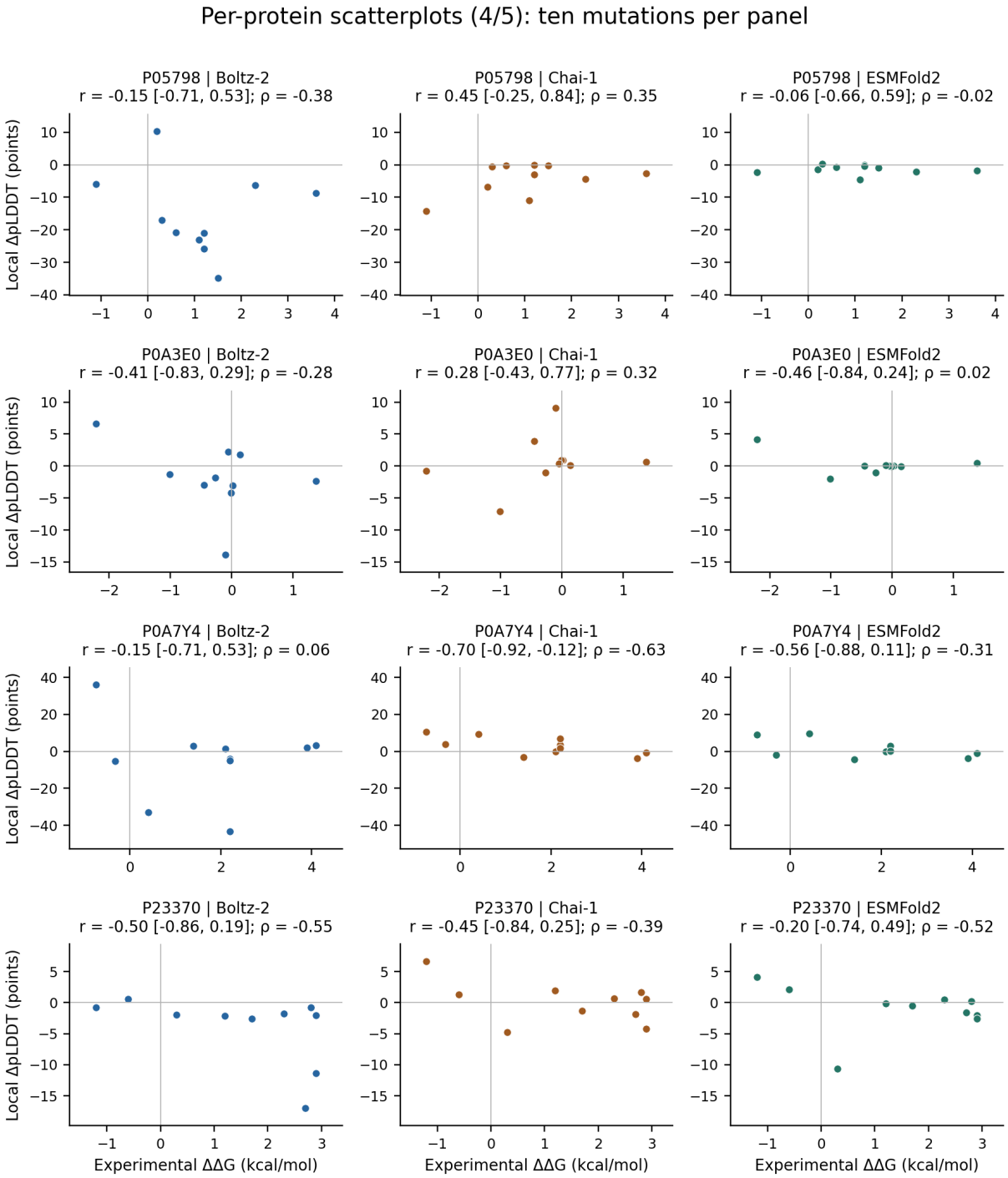


Figure S4d. Local confidence–stability scatter plots, proteins 13–16 of 20. Each panel contains all ten mutations and reports Pearson r with its approximate 95% interval and Spearman ρ. Axis ranges differ between protein rows but match across models within each row. No jitter, filtering or fitted line is applied. See Section S8 for interval assumptions.

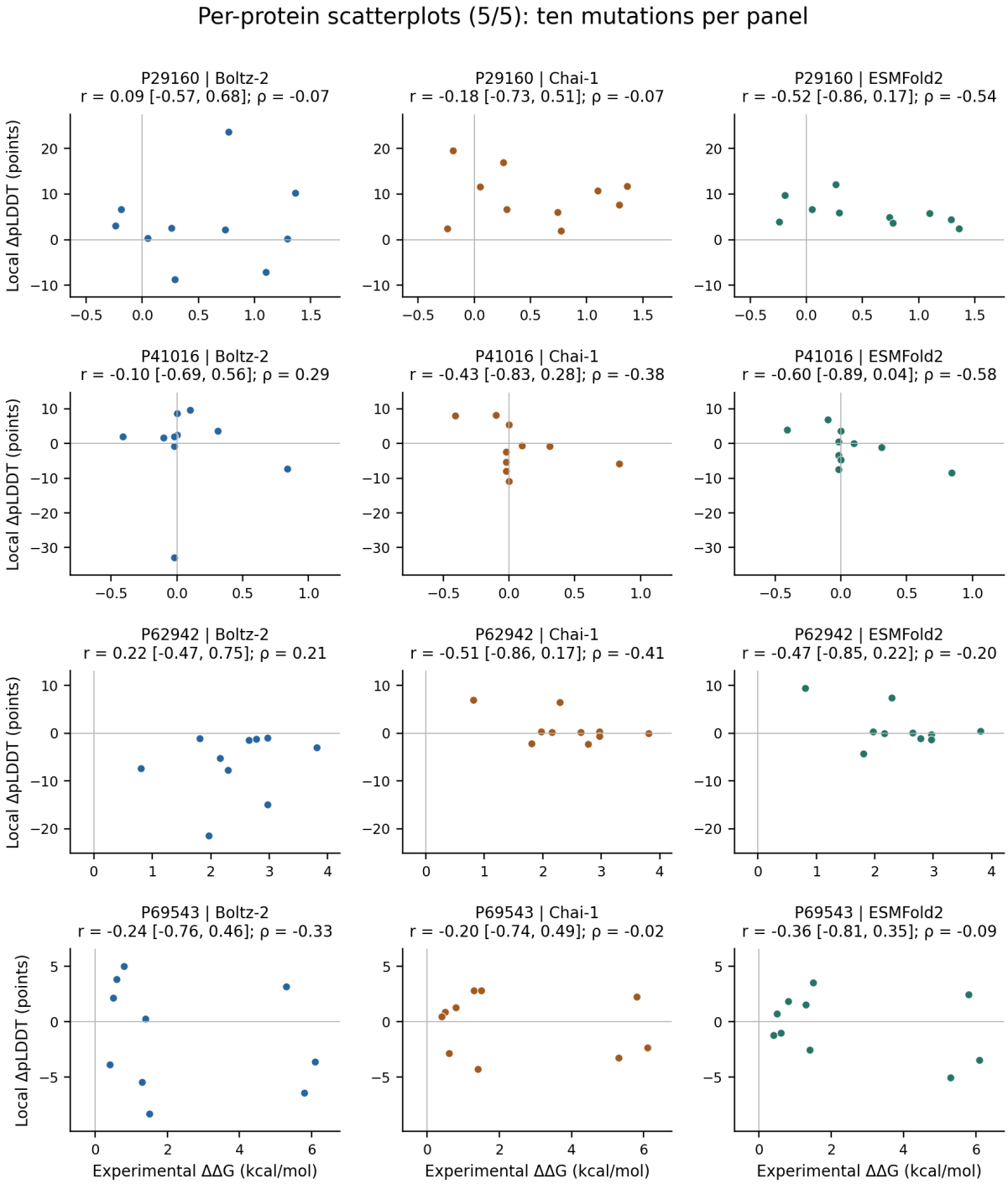


Figure S4e. Local confidence–stability scatter plots, proteins 17–20 of 20. Each panel contains all ten mutations and reports Pearson r with its approximate 95% interval and Spearman ρ. Axis ranges differ between protein rows but match across models within each row. No jitter, filtering or fitted line is applied. See Section S8 for interval assumptions.

S9 Data integrity and analysis provenance

The recovered model table contains 600 unique model–mutation rows: three models on the same 200 experimental substitutions. Experimental values, ThermoMutDB identifiers and Pak S1 row numbers agree with all independently recovered Appendix A2 records. Native-confidence verification covered 1,980 arrays and 232,551 residue values against the original completed-run snapshot. All 1,800 seed-level MT–WT comparisons and their mutation means and sample SDs were checked. No new model inference was performed during revision.

All 12 pooled coefficient estimates and protein-cluster intervals reproduce the submitted manuscript at displayed precision. All 120 per-protein model–endpoint rows, including both Pearson and Spearman coefficients, match the archived table within 5×10^{-13}. Tables and plots report rounded display values; separate unrounded exports are not included in this appendix release.

The baseline and paired-comparison analyses used shared protein-cluster bootstrap draws, as described above. The revision-analysis environment comprised Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0, pandas 2.2.3 and Matplotlib 3.10.8. These analysis-software versions are distinct from the inference environments. Source scripts that launch or install prediction models were inspected but not executed during this revision.

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